



UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	INFORMATION SYSTEMS ENGINEERING
COURSE CODE	:	ISP550/ITS572
EXAMINATION	:	JULY 2025
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
PART B (4 Questions)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

PART A

State whether each of the following statements is **TRUE** or **FALSE**.

1. Systems analysis is the process of specifying in detail how the components of an information system should be physically implemented.
2. Requirements modeling can assist development teams and stakeholders to communicate.
3. Scheduling the work is one of the activities in the 1st core process of the System Development Life Cycle (SDLC).
4. A system analyst must have capabilities of understanding how people think, learn, react to change, and must be able to motivate people.
5. In an interview, close-ended questions are usually used to encourage discussion and explanation.
6. Use cases define non-functional requirements.
7. A successful information system requires training for users, managers and IT staff members.
8. Test data should contain both correct data and erroneous data and should test all possible situations that could occur.
9. The direct cutover approach causes the changeover from the old system to the new system to occur immediately when the new system becomes operational.
10. Corrective maintenance adds enhancements to an operational system and makes the system easier to use.

(10 marks)

PART B**QUESTION 1**

- a) Briefly explain **TWO (2)** characteristics of a project for an adaptive approach to the System Development Life Cycle (SDLC)? (4 marks)
- b) Describe **THREE (3)** differences between logical and physical design of a system. (6 marks)
- c) Define the term “stakeholder” in the context of an information systems project (4 marks)
- d) Briefly explain **THREE (3)** guiding principles for Agile methodologies as expressed in The Agile Manifesto. (6 marks)

QUESTION 2

- a) Briefly explain **TWO (2)** criterias used to evaluate and rank Information System (IS) development projects. (4 marks)
- b) A system development project is considered successful when it delivers positive outcomes and is well received by its stakeholders. Explain **THREE (3)** critical factors that contribute to system development project success. (6 marks)
- c) Project feasibility helps determine whether a project is practical and worthwhile, enabling stakeholders to make informed decisions about moving forward. Assessing technical feasibility is critical during the early stages of an information systems project. Provide **THREE (3)** examples of technical feasibility assessment. (6 marks)
- d) Software application testing is a broad term that includes various types of tests. Differentiate between any **THREE (3)** types of software testing. (6 marks)

QUESTION 3

- a) Differentiate between a domain model class diagram and entity relationship diagram (ERD). (4 marks)

- b) Create a domain class diagram for online marketplace based on the following narrative.

TheBookoo.com is an online marketplace for buyers and sellers of used books. To offer books for sale, a person must register with TheBookoo.com by providing a mailing address, telephone number and e-mail address. Once registered, the system then creates and maintains an open account for this person. Sellers access to the system through a secure, authenticated portal. A seller can list books on the system using a special form that captures all the pertinent details about each book, including its category, condition, and asking price. Seller may list as many books as they wished. The system maintains an index of all books, allowing buyers to search using search engine. The search engine functionality supports queries by title, author, category, and keyword.

Buyers can browse TheBookoo.com to find and purchase books. To make a purchase, a buyer must create an account and provide credit or debit card information for payment. All account and payment data are stored on secure servers. When buyer makes a purchase, TheBookoo.com sends an email notification to the seller, including payment details, and marks the book as sold. The system keeps the order open until it receives confirmation that the book has been shipped. Upon receiving notice of a sale, the seller must notify the buyer via email within 48 hours to confirm the purchase. Shipment must be made within 24 hours after this notification. The seller then notifies both the buyer and TheBookoo.com of the shipment. Once both receive shipment notification, the system updates the order status to "shipped".

(8 marks)

- c) As a User Interface Designer, you have been tasked with creating the first version of an input interface for an online cinema ticketing system. Draw a storyboard that includes a series of sketches to represent the interaction between the user and the system based on the following sample dialog:

SYSTEM: Welcome to MovieGo! What would you like to do today?

USER: I'd like to book a movie ticket.

SYSTEM: Great! Please select the cinema and your city.

USER: TGV, Johor Bahru.

SYSTEM: Here are movies now showing in TGV Johor Bahru. Please choose one.

USER: The Marvels.

SYSTEM: Select your preferred date and time.

USER: 11 April 2025, Friday, 8:00 PM.

SYSTEM: Choose your seats (up to 5 per transaction).

USER: Seats B5 and B6.

SYSTEM: Tickets for "The Marvels" on Friday at 8:00 PM, Seats B5 and B6. Total: RM28.00. Proceed to payment?

USER: Yes.

SYSTEM: Please choose a payment method: Credit Card, Online Banking, or eWallet.

USER: Online Banking.

SYSTEM: Payment successful! Your e-ticket has been sent to your email and mobile app.

USER: Thank you.

(6 marks)

- d) Authentication and authorization are key access control mechanisms in system design that help protect data from unauthorized use. Explain **TWO (2)** potential consequences of not implementing proper access control in an information system. Provide one example for each consequence.

(6 marks)

QUESTION 4

- a) Explain the difference between a diagram and a model.

(4 marks)

- b) Nonfunctional requirements describe how the system performs under certain conditions rather than what the system does. These include aspects such as usability, reliability, performance, and supportability. Based on the nonfunctional requirement categories listed below, provide **ONE (1)** example question that you would ask a stakeholder to elicit information for each category.

Category	Example questions
i. Usability	i.
ii. Reliability	ii.
iii. Performance	iii.

(6 marks)

- c) As a system analyst, the project team has assigned you to analyse and document the system requirements for a PinkBus Company that plans to develop an Online Ticketing System. PinkBus Company aims to serve a broad range of users including daily commuters, tourists, students, and senior citizens by offering a convenient, user-friendly platform accessible via both web and mobile devices. The proposed system should allow users to view available routes, select travel destinations, choose their preferred ticket type (such as one-way, return, or day pass), and complete their purchase using a variety of payment methods. These include Touch 'n Go eWallet, credit or debit cards, online banking, and QR code payments from e-wallet services like GrabPay and Boost. To accommodate tourists and users with different needs, the system should support multiple languages (e.g., English and Malay) and offer accessibility features such as larger font options and simplified layouts. After a successful transaction, the system will generate and deliver a digital ticket, which users can display on their device when boarding the bus. These tickets should be sent via email or in application notifications. The system should also log all transactions for audit and reporting purposes. In case of payment failures, ticket issues, or technical problems, users should have access to a customer support channel, while system administrators should be able to manage schedules, update fare tables, monitor usage, and perform routine maintenance. Draw a Unified Modelling Language (UML) use case diagram for the above narrative.

(8 marks)

- d) System implementation is the critical phase in the system development life cycle where the designed system is installed, tested, and deployed for actual use. Despite careful planning, many implementations fail or fall short of expectations due to several key reasons. Explain **THREE (3)** key reasons why system implementation may fail. Support your explanation with relevant example.

(6 marks)

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END OF QUESTION PAPER